

# Relative price dispersion and inflation regimes in South Africa

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**JEL Codes:**

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# 1 Introduction

Inflation is known to have adverse welfare effects through its distortion of relative prices which causes the misallocation of resources, production inefficiencies leading to lower real economic activity. This is well supported by both theoretical and empirical studies. This paper contributes to the empirical literature, by examining the relationship between inflation and relative price variability (RPD) in South Africa given the little evidence available and to better understand how this relationship has evolved and the implications for monetary policy.

The South African context can provide further insights into the relationship between inflation and RPD given the shifts in the inflation targets from 3-6 per cent in 2000 to 4.5 per cent since 2017 as structural changes in inflation can affect the link between inflation and relative price variability (Caglayan and Filiztekin, 2003) and that the functional form of relationship is U-shaped and dynamic (Fielding and Mizen, 2000; Choi, 2010). The latter motivates the estimation of the inflation rate that minimizes relative price distortions since if the relationship was linear then the optimal inflation rate would be zero. Given the inflation target reform towards a 3% in South Africa, this study contributes to the policy discussion on the appropriate inflation target and understating how dynamic relations between inflation and relative price variability has evolved.

The theoretical literature generally distinguishes between menu-cost and signal extraction models in explaining the positive relation between trend inflation and RPD. In these models' firms follow a one-sided dynamic pricing rule when setting prices and face fixed non-zero costs of adjusting nominal prices (e.g. printing new menus) when inflation increases as the latter reduces the optimal real price of goods and services. Due to differences in administrative costs, these nominal price adjustment costs vary across firms moreover these adjustments occur at different times which implies staggered price setting behaviour which leads to relative price variability. Menu cost models predict a that higher expected inflation causes an increase in RPD (citations xxx) while Lucas-type signal extraction models (Lucas, 1973; Hercowitz, 1981; Cukierman, 1983) argue unexpected inflation uncertainty causes higher RPD because informational frictions make it difficult to distinguish between idiosyncratic relative shocks or

an aggregate (inflationary) shock. Menu cost and signal-extraction models assume a positive and linear relationship making the optimal inflation rate zero, however, recently monetary search models (Head and Kumar, 2005) show that this relationship is nonlinear making the optimal inflation rate somewhere between zero and positive inflation rate.

## 2 Citations

Adam et al. (2024) Baglan et al. (2016) Ball and Romer (1993) Benabou and Gertner (1993) Caglayan and Filiztekin (2003) Caglayan et al. (2008) Grier and Perry (1996) Lach and Tsiddon (1992) Nakamura et al. (2018) Ndou and Redford (2014) Ndou and Gumata (2017) Parks (1978) Parsley (1996) Rather et al. (2018)

## 3 Literature review

## 4 Data and methodology

Our analysis uses monthly inflation data on 45 goods categories from January 2009 to NoV 2025 obtained from Statistics South Africa (Stats SA).<sup>1</sup> . Figure A1 below shows the long term inflation rate in South Africa, and Figure A2 in the appendix shows the various price categories used in the analysis. Table A1 provides the descriptive statistics of the inflation rates for the various price categories used in the analysis.

The relative price dispersion (RPD) is calculated in line with Lach and Tsiddon (1992) as the weighted standard deviation of individual price inflation rates from the aggregate inflation rate. The RPD is essentially the variability of relative prices of a given product. We first calculate the monthly inflation rate as follows:

$$\pi_{i,t} = \ln \left( \frac{P_{i,t}}{P_{i,t-1}} \right) * 100, \quad (1)$$

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<sup>1</sup>The data is available at: <http://www.statssa.gov.za/> The data is seasonally adjusted.

where  $\pi_{i,t}$  is the inflation rate of good  $i$  at time  $t$ ,  $P_{i,t}$  is the price of good  $i$  at time  $t$ , and  $P_{i,t-1}$  is the price of good  $i$  at time  $t-1$ . Then, the RPD is calculated as:

$$RPD_t = \sqrt{\sum_{i=1}^n \omega_i (\pi_{i,t} - \pi_t)^2}, \quad (2)$$

where  $\omega_j$  is the expenditure share of the  $j^{th}$  good,  $\pi_t$  is the headline or aggregate inflation rate at time  $t$ , and  $\sum_{i=1}^n \omega_i = 1$ . The expenditure weights ( $\omega_i$ ) used are derived from the Consumer Price Index (CPI) basket provided by Stats SA. Figure 1 shows the RPD calculated using the above formula. We use the 2025 CPI Stats SA weights. The weights are shown in Figure A3 in the appendix.

To examine the relationship between inflation and RPD, we estimate the following econometric model:

$$RPD_t = \beta_0 + \beta_1 |\pi_t| + \sum_{h=1}^3 \omega_h RPD_{t-h} + \mu_t, \quad (3)$$

where  $RPD_t$  is the relative price variance at time  $t$ ,  $|\pi_t|$  is the absolute value of aggregate inflation rate at time  $t$ ,  $RPD_{t-h}$  are lagged values of RPD to account for its dynamic nature, and  $\mu_t$  is the error term. The coefficient  $\beta_1$  captures the effect of inflation on RPD. We include three lags of RPD based on Akaike Information Criterion (AIC) (and other tests) and to capture potential persistence in RPD. For robustness, we repeat this estimation using core inflation (excluding food and non-alcoholic beverages, and fuel and electricity) instead of headline inflation to account for the potential volatility in these components.

The literature on RPD emphasises that the relationship between inflation and RPD may be time-varying due to structural changes in the economy, monetary policy regimes, and other factors. To capture potential time variation in the relationship, we also estimate a rolling regression model with a 24-month and 60-month windows as follows:

$$RPD_t = \alpha_t + \beta_{1,t} |\pi_t| + \sum_{h=1}^3 \omega_{h,t} RPD_{t-h} + \mu_t, \quad (4)$$

where the coefficients  $\alpha_t$ ,  $\beta_{1,t}$ , and  $\omega_{h,t}$  vary over time. This approach allows us to observe the

stability of the coefficients over time, which is an indicator of non-linearity in the relationship between RPD and inflation.

Next, we utilise a semi-parametric or partially linear approach to estimate the optimal level of inflation given the RPD. This approach allows us to model the relationship between inflation and RPD without imposing a strict functional form. Following the literature, we first specify a quadratic (U-shaped) relationship between inflation and RPD as follows:

$$RPD_t = \alpha + \pi_t + \beta_2 \pi_t^2 + \sum_{h=1}^3 \phi_h RPD_{t-h} + \epsilon_t, \quad (5)$$

where  $\beta_2$  captures the curvature of the relationship between inflation and RPD. The optimal inflation rate that minimizes RPD can be derived by taking the first derivative of the equation with respect to  $\pi_t$  and setting it to zero, yielding:

$$\pi_t^* = -\frac{1}{2\beta_2}. \quad (6)$$

To estimate this optimal inflation we implement a semi parametric approach as follows:

$$RPD_t = f(\pi_t) + \sum_{h=1}^3 \phi_h RPD_{t-h} + \epsilon_t, \quad (7)$$

where  $f(\pi_t)$  is a non-linear smooth differentiable function of inflation and the RPD. Therefore, the optimal inflation rate can be obtained by finding the minimum of the estimated  $f(\pi_t)$  function.  $f(\pi_t)$  is estimated in two stages. First we estimate  $\lambda_h$  as follows:

$$RPD_t = \sum_{h=1}^3 \lambda_h \overline{RPD}_{t-h} + \eta_t, \quad (8)$$

where  $\overline{RPD}_{t-h}$  are the residuals from a non-parametric regression of each lag of  $RPD_t$  on  $\pi_t$ . In the second stage, we estimate  $f(\pi_t)$  non-parametrically as follows:

$$\hat{\eta}_t = f(\pi_t) + v_t, \quad (9)$$

where  $\hat{\eta}_t$  are the residuals from Equation 8, and  $f(\pi_t)$  is estimated using a kernel regression

method. The bandwidth ( $h$ ) for the kernel estimator is selected using cross-validation to minimize the mean squared error. As a treatment for extreme values of inflation, we further implement the unbounded Gaussian kernel and the outlier-robust Epanechnikov kernel. In essence, the semi-parametric approach captures the sensitivity of RPD to marginal increases in inflation. In that case, the optimal inflation rate can be obtained by finding the point where the first derivative ( $f'(\pi_t)$ ) is equal to zero. Therefore, if  $f'(\pi_t) > 0$  ( $f'(\pi_t) < 0$ ), then an increase in inflation (decrease) leads to an increase (decrease) in the RPD. For robustness we estimate the  $g(UIN_t)$  with multiple bandwidth ( $h$ ).

To further disentangle the effects of RPD on inflation, we decompose headline inflation into its expected and unexpected components. As discussed in the literature section, for example, theory suggests that inflation uncertainty has a more pronounced effect on RPD due to information frictions faced by firms when adjusting prices (Lucas, 1973; Hercowitz, 1981; Cukierman, 1983).

We therefore estimate the following model:

$$RPD_t = \alpha + \beta_0 EIN_t + \beta_1 UIN_t + g(UIN_t) + \sum_{k=1}^3 \lambda_k RPD_{t-k} + \epsilon_t, \quad (10)$$

where  $EIN_t$  is the expected inflation rate,  $UIN_t$  is the unexpected inflation rate, and  $g(UIN_t)$  is a non-linear function of unexpected inflation ( $UIN_t$ ). We obtain the expected using **a simple moving average of headline inflation, and unexpected inflation as the standard deviation of inflation**. The estimation of  $g(UIN_t)$  follows the same two-stage semi-parametric approach outlined in Equation 8 and Equation 9. The number of lags is the same as in Equation 3.

Lastly, and some-what novel to the RPD literature, we employ a quantile on quantile regression approach (Koenker, 2017). Quantile on quantile regressions are robust to outliers and allows us to estimate the regime aspects of the relationship between RPD and inflation. For brevity, we explain the quantile on quantile estimation procedure in the Appendix.

## 5 Results

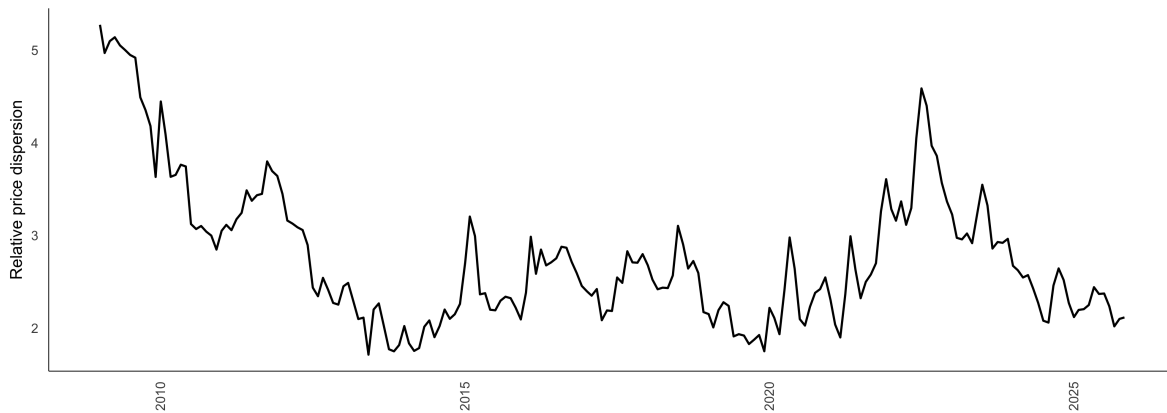


Figure 1: RPD over-time

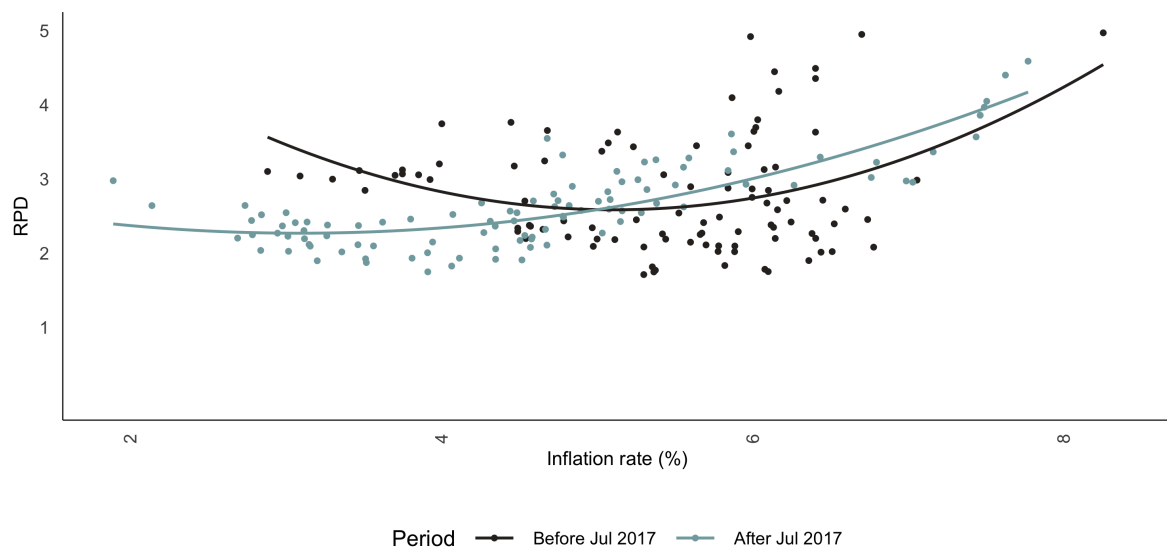


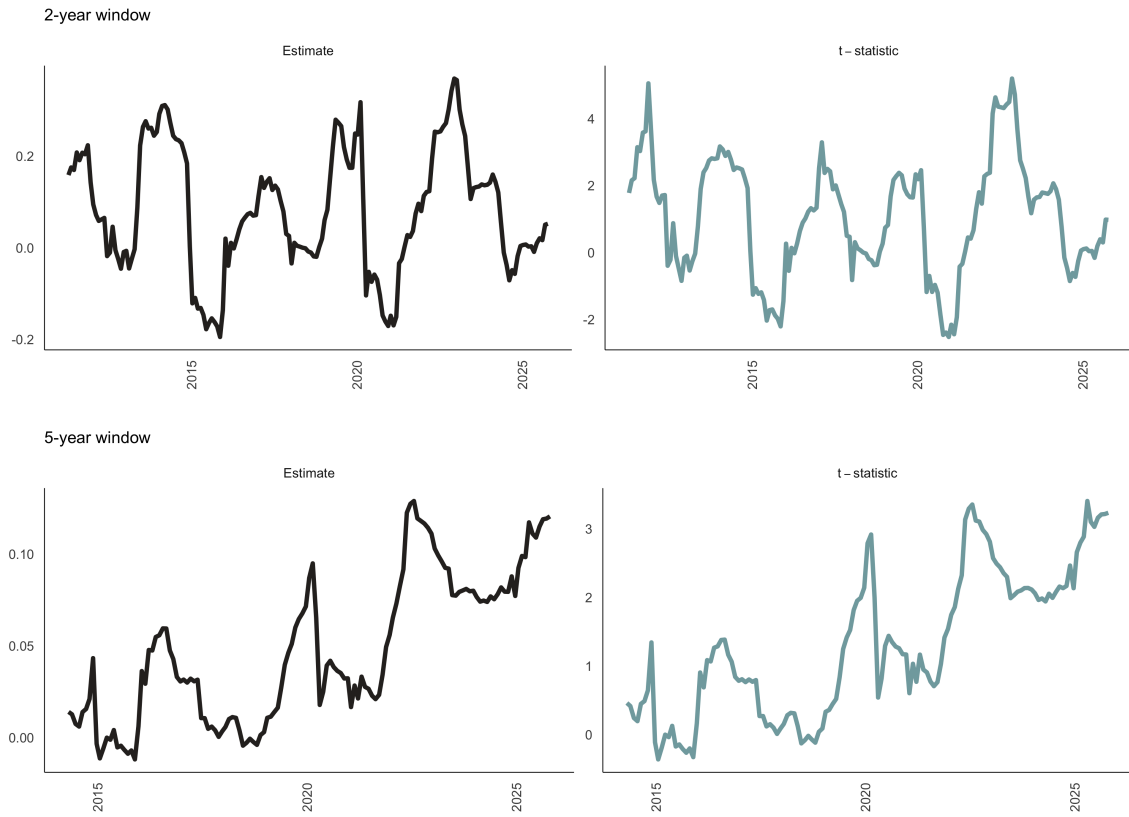
Figure 2: RPD vs headline Inflation

Table 1: Baseline regressions

|           | Full Sample | Pre 2017 | Post 2017 | Full Sample: Squared Inflation | Pre 2017: Squared Inflation | Post 2017: Squared Inflation |
|-----------|-------------|----------|-----------|--------------------------------|-----------------------------|------------------------------|
| $\pi_t$   | 0.01*       | 0.009    | 0.032***  | -0.046                         | -0.104                      | -0.033                       |
| $\pi_t^2$ |             |          |           | 0.006*                         | 0.011*                      | 0.007*                       |

|                  | Full Sample | Pre 2017 | Post 2017 | Full Sample: Squared Inflation | Pre 2017: Squared Inflation | Post 2017: Squared Inflation |
|------------------|-------------|----------|-----------|--------------------------------|-----------------------------|------------------------------|
| $RPD_{\Delta-3}$ | 0.244***    | 0.173*   | 0.248**   | 0.234***                       | 0.154                       | 0.209**                      |
| $RPD_{\Delta-2}$ | -0.382***   | -0.14    | -0.645*** | -0.363***                      | -0.111                      | -0.609***                    |
| $RPD_{\Delta-1}$ | 1.039***    | 0.896*** | 1.116***  | 1.017***                       | 0.864***                    | 1.063***                     |
| $D_{2022m8}$     | 0.026       |          | 0.024     | -0.001                         |                             | -0.012                       |
| $D_{2020m2}$     | -0.116      |          | -0.213**  | -0.106                         |                             | -0.205**                     |
| $D_{2017m7}$     | 0.156*      |          |           | 0.161*                         |                             |                              |
| $D_{2014m7}$     | -0.098      | -0.087   |           | -0.105                         | -0.096                      |                              |
| $D_{2011m12}$    | 0.042       | 0.027    |           | 0.047                          | 0.037                       |                              |
| $C$              | 0.044       | 0.012    | 0.117***  | 0.184**                        | 0.326*                      | 0.304**                      |

Note: \*\*\* = p value < 0.01, \*\* = 0.01 < p value < 0.05, \* = 0.05 < p value < 0.1.



Note: t-statistics of greater (or less) than 1.96 ( or -1.96) is significant at a 5% level of significance.

Figure 3: Rolling regressions



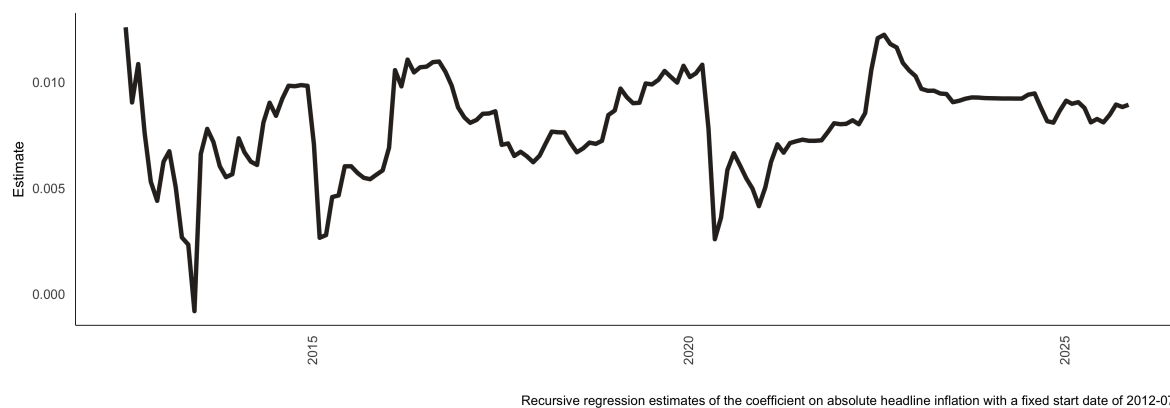


Figure 4: Recursive regressions

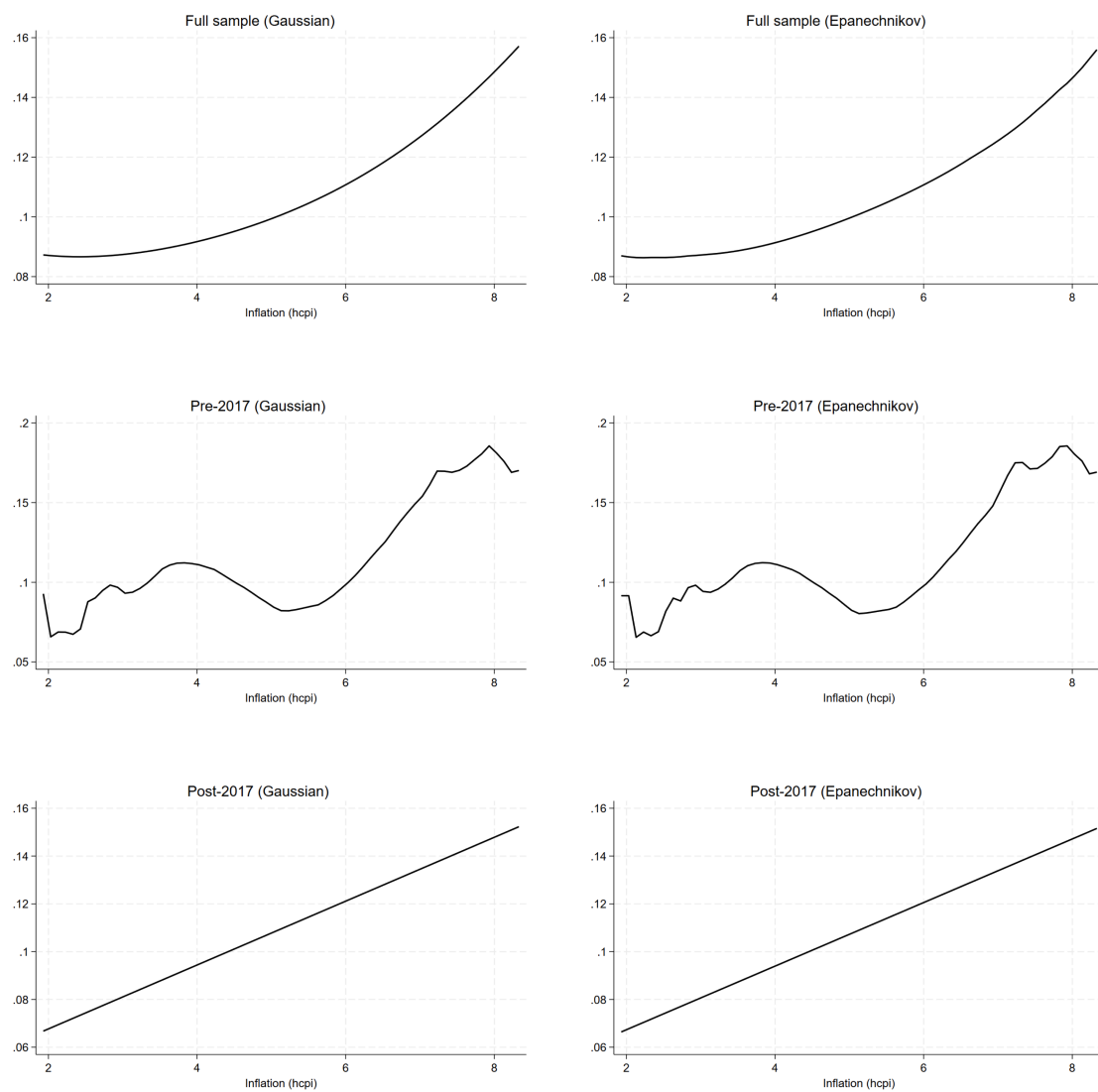


Figure 5: Semi-parametric regressions

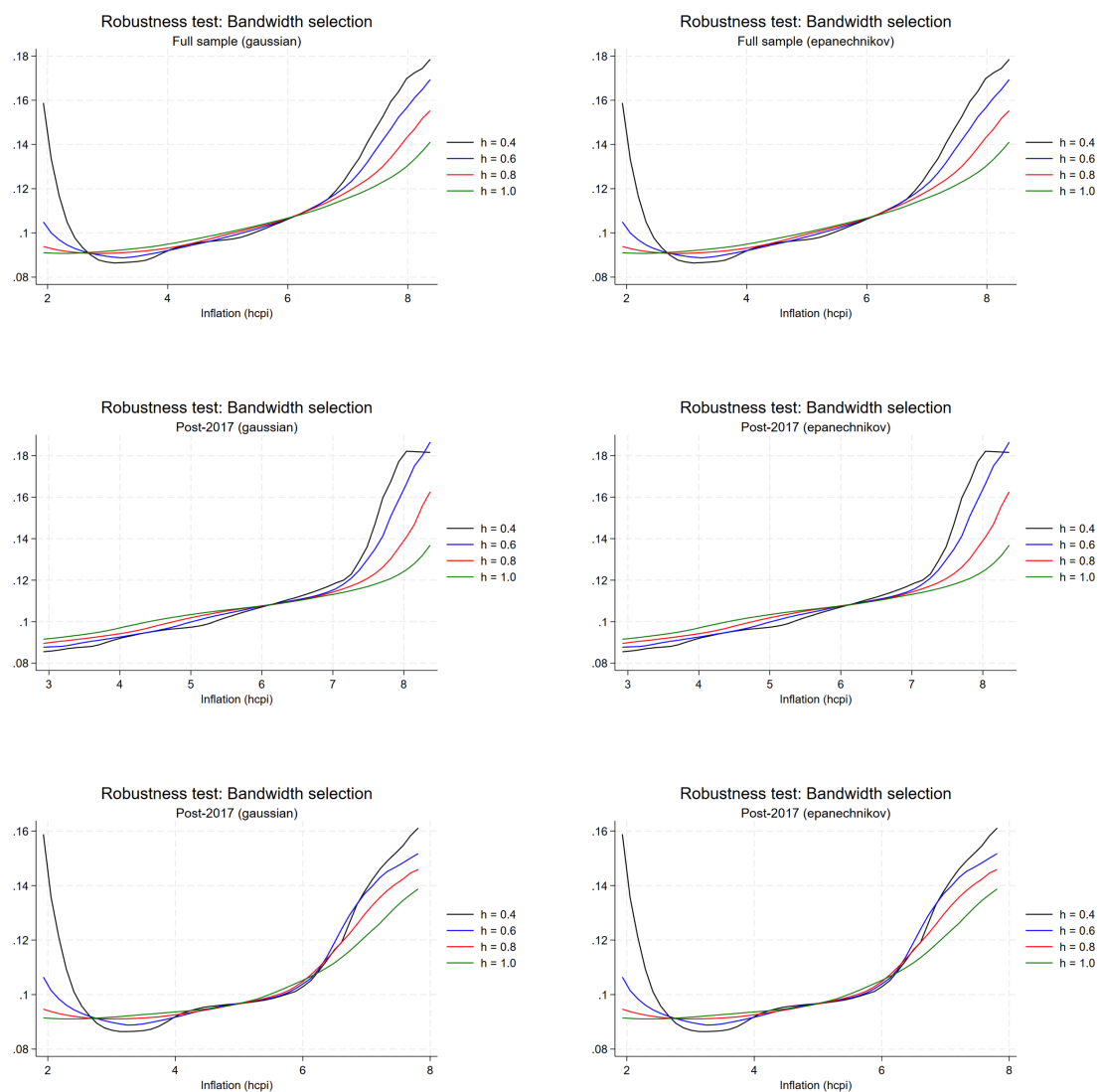


Figure 6: Semi-parametric regressions with alternative bandwidth selection

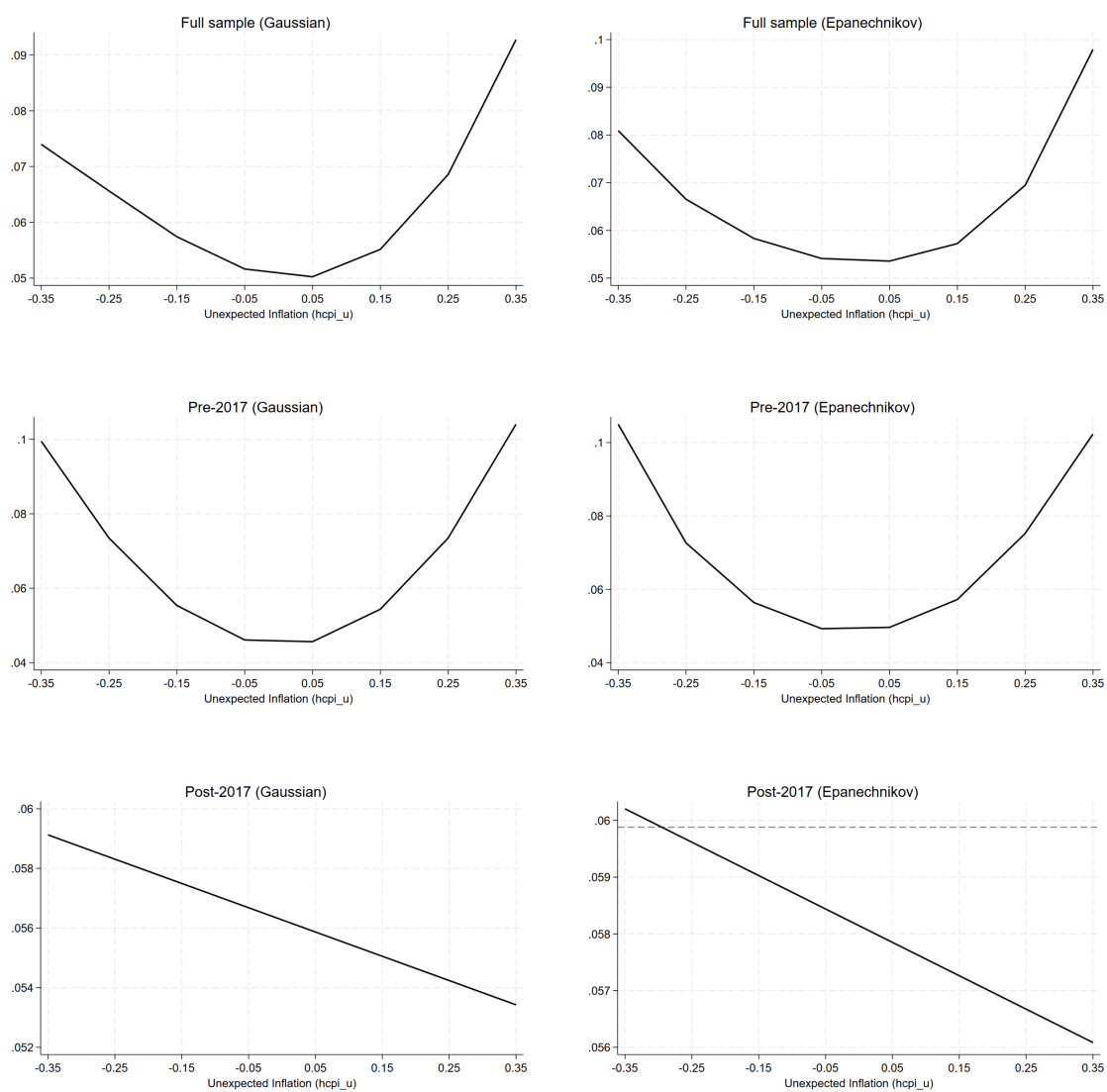


Figure 7: Semi-parametric regressions for unexpected inflation

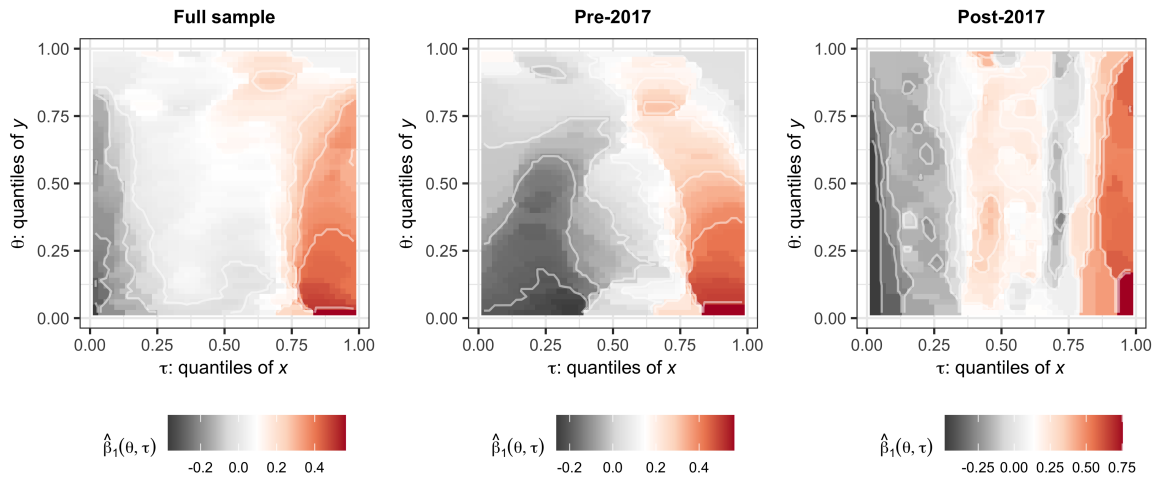


Figure 8: Quantile on quantile regressions

## 6 Conclusion

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## A Appendix: Data and descriptive statistics

### A.1 Data

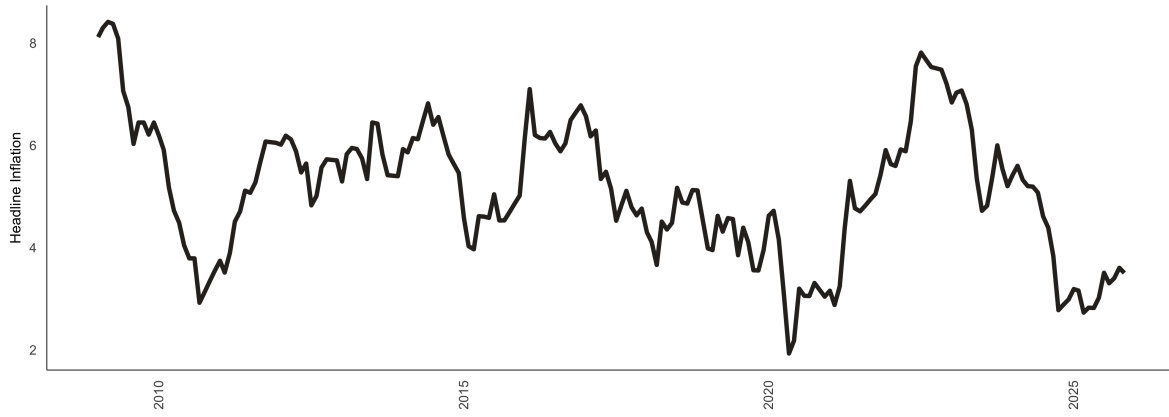


Figure A1: Headline inflation (%).

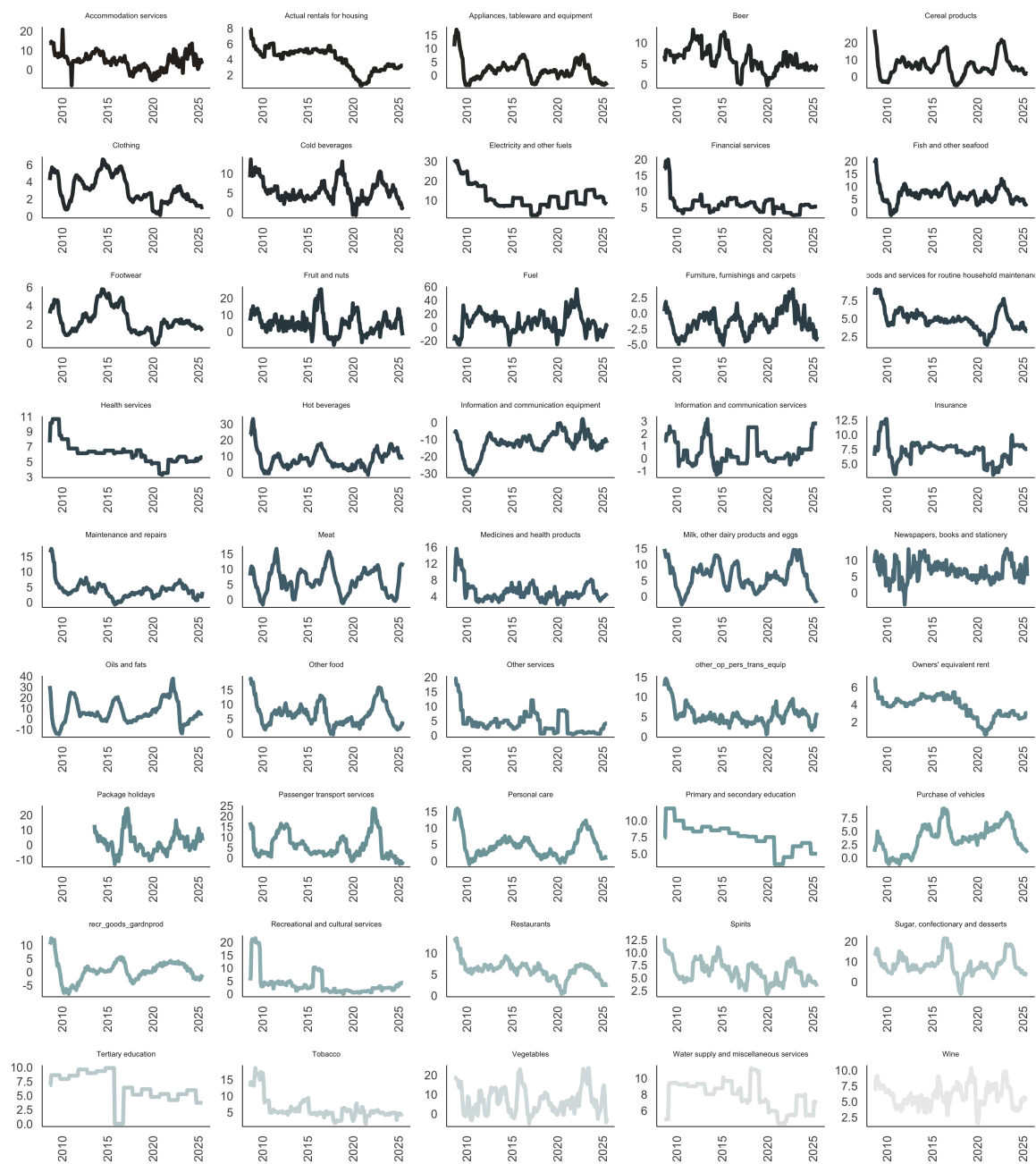


Figure A2: Inflation categories (%)

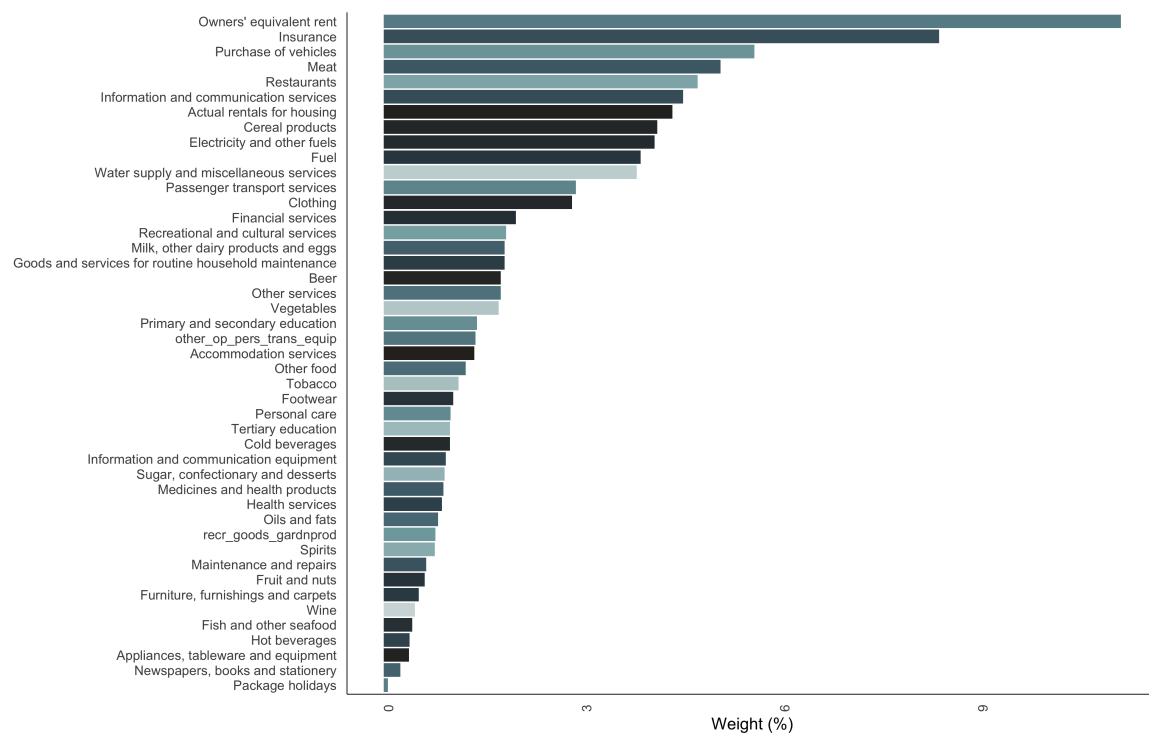


Figure A3: CPI Weights. Note: Due to a lack of data, some categories (no spirits; transport services of goods; household textiles; glassware, tableware, and house utensils; and Tools and equipment for house and garden) have been excluded.

## A.2 Descriptive statistics

Table A1: Descriptive statistics

| Goods category                                       | Mean   | Median | SD    | Min    | Max   | Observations |
|------------------------------------------------------|--------|--------|-------|--------|-------|--------------|
| Accommodation services                               | 4.47   | 4.55   | 4.43  | -8.16  | 20.79 | 203          |
| Actual rentals for housing                           | 4.03   | 4.56   | 1.48  | 0.67   | 7.80  | 203          |
| Appliances, tableware and equipment                  | 1.81   | 1.50   | 3.93  | -3.75  | 16.97 | 203          |
| Beer                                                 | 6.07   | 5.72   | 2.85  | -0.25  | 13.19 | 203          |
| Cereal products                                      | 6.19   | 5.17   | 6.17  | -5.18  | 27.78 | 203          |
| Clothing                                             | 3.06   | 2.68   | 1.63  | 0.11   | 6.60  | 203          |
| Cold beverages                                       | 5.51   | 5.02   | 2.69  | -0.76  | 13.59 | 203          |
| Electricity and other fuels                          | 11.78  | 10.66  | 6.19  | 2.28   | 30.99 | 203          |
| Financial services                                   | 5.82   | 5.43   | 2.96  | 2.51   | 20.00 | 203          |
| Fish and other seafood                               | 6.54   | 6.43   | 3.22  | -1.47  | 20.74 | 203          |
| Footwear                                             | 2.45   | 2.16   | 1.40  | -0.22  | 5.77  | 203          |
| Fruit and nuts                                       | 4.75   | 4.25   | 6.16  | -8.18  | 25.36 | 203          |
| Fuel                                                 | 6.51   | 7.33   | 15.01 | -26.78 | 56.28 | 203          |
| Furniture, furnishings and carpets                   | -1.42  | -1.43  | 1.85  | -5.11  | 3.83  | 203          |
| Goods and services for routine household maintenance | 4.88   | 4.76   | 1.44  | 1.32   | 9.03  | 203          |
| Health services                                      | 6.12   | 6.11   | 1.48  | 3.26   | 10.65 | 203          |
| Hot beverages                                        | 8.16   | 6.90   | 5.98  | -1.70  | 33.08 | 203          |
| Information and communication equipment              | -12.55 | -11.75 | 6.61  | -30.92 | 1.93  | 203          |
| Information and communication services               | 0.56   | 0.20   | 1.05  | -1.32  | 3.15  | 203          |
| Insurance                                            | 7.19   | 7.51   | 1.81  | 3.14   | 12.68 | 203          |
| Maintenance and repairs                              | 4.09   | 3.85   | 2.84  | -0.80  | 17.53 | 203          |
| Meat                                                 | 6.20   | 6.03   | 4.16  | -1.67  | 16.67 | 203          |
| Medicines and health products                        | 5.05   | 4.46   | 2.31  | 1.97   | 15.53 | 203          |
| Milk, other dairy products and eggs                  | 6.07   | 6.12   | 4.16  | -2.60  | 14.83 | 203          |
| Newspapers, books and stationery                     | 6.81   | 6.56   | 3.03  | -3.72  | 13.50 | 203          |
| Oils and fats                                        | 6.17   | 4.21   | 10.55 | -14.55 | 37.67 | 203          |
| Other food                                           | 6.56   | 6.11   | 4.00  | -0.45  | 18.83 | 203          |
| Other services                                       | 4.28   | 3.40   | 3.83  | 0.31   | 19.47 | 203          |
| Owners' equivalent rent                              | 3.69   | 3.92   | 1.25  | 0.56   | 6.98  | 203          |
| Package holidays                                     | 1.14   | 0.36   | 7.23  | -13.17 | 24.37 | 203          |
| Passenger transport services                         | 5.67   | 3.15   | 5.75  | -3.09  | 23.67 | 203          |
| Personal care                                        | 4.51   | 3.86   | 3.68  | -1.07  | 15.93 | 203          |
| Primary and secondary education                      | 7.60   | 7.64   | 1.98  | 3.42   | 11.80 | 203          |
| Purchase of vehicles                                 | 3.66   | 3.80   | 2.60  | -1.23  | 9.28  | 203          |
| Recreational and cultural services                   | 3.77   | 2.24   | 4.64  | -0.11  | 21.58 | 203          |
| Restaurants                                          | 6.03   | 6.18   | 2.16  | 0.38   | 13.50 | 203          |
| Spirits                                              | 6.15   | 6.04   | 2.14  | 1.81   | 12.79 | 203          |
| Sugar, confectionary and desserts                    | 8.51   | 7.80   | 5.06  | -6.03  | 21.61 | 203          |
| Tertiary education                                   | 6.48   | 6.11   | 2.52  | 0.00   | 9.85  | 203          |
| Tobacco                                              | 6.04   | 4.93   | 3.44  | 1.40   | 18.58 | 203          |
| Vegetables                                           | 6.89   | 5.72   | 6.65  | -5.14  | 23.59 | 203          |
| Water supply and miscellaneous services              | 7.77   | 8.02   | 1.79  | 4.38   | 11.23 | 203          |
| Wine                                                 | 5.97   | 5.93   | 1.68  | 1.47   | 10.35 | 203          |
| coreinfl                                             | 4.57   | 4.54   | 1.13  | 2.50   | 8.46  | 203          |
| hinfl                                                | 5.13   | 5.17   | 1.32  | 1.93   | 8.41  | 203          |
| other_op_pers_trans equip                            | 5.37   | 4.94   | 2.44  | 0.52   | 14.60 | 203          |
| recr_goods_gardnprod                                 | 0.42   | 0.39   | 3.83  | -8.23  | 12.51 | 203          |

### A.3 Stationarity tests

Table A2: Stationarity tests results

| Measure            | ADF test (p-value) | PP test (p-value) | KPSS test (p-value) | Stationarity |
|--------------------|--------------------|-------------------|---------------------|--------------|
| Headline Inflation | 0.14               | 0.20              | 0.04                | Stationary   |
| RPD                | 0.08               | 0.31              | 0.01                | Stationary   |

### A.4 Lag length selection

Table A3: RPD lag length selection

| Tests  | Optimal Lags |
|--------|--------------|
| AIC(n) | 3            |
| HQ(n)  | 3            |
| SC(n)  | 1            |
| FPE(n) | 3            |

## B Appendix: Alternative Results (OLS Robustness)

The coefficient of variation (CV) version of RPD (or RPD-CV) normalizes the RPD by the aggregate price level to account for changes in the overall price level. This measure is particularly useful when comparing RPD across different time periods or economies with varying inflation rates. The RPD-CV is calculated as follows:

$$RPD - CV_t = \sqrt{\frac{\sum_{i=1}^n \omega_i (\pi_{i,t} - \pi_t)^2}{|1 + \pi_t|}}, \quad (11)$$

where  $\pi_t$  is the aggregate inflation rate at time  $t$ , and  $\omega_i$  are the inflation basket weights.

### B.1 RPD-CV over-time

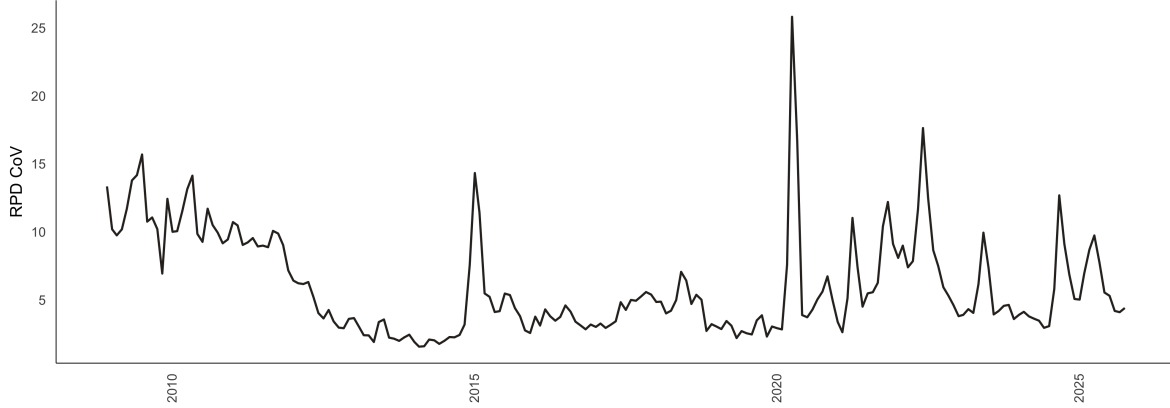


Figure B1: RPD-CV overtime

## B.2 Stationarity tests for RPD-CV

Table B1: Stationarity tests results for RPD-CV

| Measure            | ADF test (p-value) | PP test (p-value) | KPSS test (p-value) | Stationarity |
|--------------------|--------------------|-------------------|---------------------|--------------|
| Headline Inflation | 0.14               | 0.20              | 0.04                | Stationary   |
| RPD-CV             | 0.15               | 0.01              | 0.02                | Stationary   |

## B.3 Lag length selection for RPD-CV

Table B2: RPD-CV lag length selection

| Tests  | Optimal Lags |
|--------|--------------|
| AIC(n) | 3            |
| HQ(n)  | 3            |
| SC(n)  | 3            |
| FPE(n) | 3            |



## B.4 RPD-CV baseline regression results

Table B3: RPD-CV baseline regressions

|                 | Full Sample | Pre 2017 | Post 2017 | Full Sample: Squared Inflation | Pre 2017: Squared Inflation | Post 2017: Squared Inflation |
|-----------------|-------------|----------|-----------|--------------------------------|-----------------------------|------------------------------|
| $\pi_t$         | -0.014      | -0.006   | 0.001     | -0.242**                       | -0.342*                     | -0.383**                     |
| $\pi_t^2$       |             |          |           | 0.023**                        | 0.031*                      | 0.04***                      |
| $RPD_{t-3}$     | 0.348***    | 0.154    | 0.361***  | 0.329***                       | 0.126                       | 0.286***                     |
| $RPD_{t-2}$     | -0.648***   | -0.298*  | -0.792*** | -0.618***                      | -0.259*                     | -0.725***                    |
| $RPD_{t-1}$     | 1.18***     | 1.077*** | 1.171***  | 1.145***                       | 1.036***                    | 1.065***                     |
| $D_{2022m8}$    | -0.144*     |          | -0.074    | -0.235                         |                             | -0.212                       |
| $D_{2020m2}$    | -0.371***   |          | -0.498*   | -0.338                         |                             | -0.45                        |
| $D_{2017m7}$    | 0.295***    |          |           | 0.314                          |                             |                              |
| $D_{2014m7}$    | -0.125**    | -0.139   |           | -0.159                         | -0.17                       |                              |
| $D_{2011m12}$   | 0.056       | -0.009   |           | 0.083                          | 0.028                       |                              |
| <i>Constant</i> | 0.261**     | 0.132    | 0.43***   | 0.844***                       | 1.066**                     | 1.475***                     |

Note: \*\*\* = p value < 0.01, \*\* = 0.01 < p value < 0.05, \* = 0.05 < p value < 0.1.

## C Appendix: Explanation of the quantile on quantile regression

To further capture the dynamic dependencies in the data, we perform a quantile on quantile regressions. Quantile regressions address the limitations of OLS which only estimates the mean dependency between one or more independent variables and the dependant variable. Distribution moments, such as the mean, can be strongly affected by heavy tails (or tail behaviour), non-linearity and extreme values in general (see [Koenker, 2017](#)). In response to these limitations, the quantile regression was first introduced by [Koenker and Bassett Jr \(1978\)](#). [Koenker \(2017\)](#) notes that quantile regressions are inherently local and are immune to small deviations in distributions.

However, [Gupta et al. \(2018\)](#) note that standard quantile regressions are limited in their ability to capture dependency in its entirety. That is, although quantile regressions capture the relationship between two variable (for example) at various points of the conditional distribution, it restricts the possibility that the nature of the independent variable can also influence how the independent variable is calculated. The quantile-on-quantile regression (QQR), therefore, offers a more complete picture of the complex dependency structure, and was utilised numerous times in the literature (see [Mishra et al., 2019](#); [Chang et al., 2020](#); [Sim, 2016](#)).

Given this brief explanation on QQR, we estimate the relationship between the  $\theta$ -quantile of RPD ( $a_{2t}$ ) and the  $\tau$ -quantile of headline inflation ( $\Gamma_t$ ) as follows:

$$a_{2t} = \beta^\theta \Gamma_t + \epsilon_t^\theta, \quad (12)$$

Taking the linear version of  $\beta^\theta(\cdot)$  with a first order Taylor expansion of  $\beta^\theta(\cdot)$  around  $\Gamma^\tau$  gives:

$$\beta^\theta(\Gamma_t) \approx \beta^\theta(\Gamma^\tau) + \beta^{\theta'}(\Gamma^\tau)(\Gamma_t - \Gamma^\tau) \quad (13)$$

In this instance both  $\beta^\theta(\Gamma^\tau)$  and  $\beta^{\theta'}(\Gamma^\tau)$  are indexed to  $\theta$  and  $\tau$ . This means that Equation 13 can be summarised as follows:

$$\beta^\theta(\Gamma_t) \approx \beta_0(\theta, \tau) + \beta_1(\theta, \tau)(\Gamma_t - \Gamma), \quad (14)$$

and collecting terms gives and substituting into Equation 12 results in :

$$a_{2t} = \beta_0(\theta, \tau) + \beta_1(\theta, \tau)(\Gamma^\tau)(\Gamma_t - \Gamma) + \epsilon_t^\theta \quad (15)$$

where  $\beta^\theta(\Gamma^\tau)$  and  $\beta^{\theta'}(\Gamma^\tau)$  are substituted with  $\beta_0(\theta, \tau)$  and  $\beta_1(\theta, \tau)(\Gamma^\tau)$  respectively.

Therefore, Equation 15 captures the relationship between the  $\theta$ -quantile of the RPD and the  $\tau$ -quantile of inflation, that is, the overall dependence structure of the respective distributions.

To this end, to get the estimates of  $\hat{\beta}_0(\theta, \tau)$  and  $\hat{\beta}_1(\theta, \tau)$  we solve for:

$$\min_{b_0, b_1} \sum_{i=1}^n \rho_\theta [a_{2t} - b_0 - b_1(\widehat{\Gamma}_t - \hat{\Gamma})] K \left( \frac{F_n(\widehat{\Gamma}_t) - \tau}{h} \right) \quad (16)$$

where  $\rho_\theta$  is a tilted absolute value function which produces  $\theta$ -quantile values of  $a_{2t}$ . A Gaussian kernel  $K(\cdot)$  is used to weight observations in the neighbourhood of  $\widehat{\Gamma}^\tau$  using  $h$  as a bandwidth. These weights are inversely related to  $\widehat{\Gamma}_t - \hat{\Gamma}$ . Lastly, the choice of  $h$  remains uncertain in kernel regression where if a small  $h$  is chosen the bias of the estimates is smaller but the variance of these estimates increases, and conversely. To overcome this uncertainty we use the cross validation method to calculate the appropriate level of  $h$ .